Heat-flux affinity sweep and Gaussian crossover

Frozen $n = 10$ Cartesian experiment with auditable bootstrap-gate amendment, 2026-09-04

Scope and frozen estimator

The transported heat in a window is

$$Q = \frac{Q_L - Q_R}{2}, \qquad \Delta\beta = \frac{1}{T_R} - \frac{1}{T_L}.$$

Non-overlapping windows are formed within each of 128 independent streams. The directly sampled fluctuation-relation slope is obtained from raw matched bins of $\log[p(Q)/p(-Q)]$ against Q. The fixed rule is $\Delta Q = \mathrm{std}(Q)/20$, at least 10 raw observations on each side of every accepted bin pair, and the largest contiguous reliable block that straddles zero. No plus-four points or extrapolated densities enter the fit. Stream bootstrap intervals resample the 128 complete streams.

The Gaussian bulk diagnostics are

$$a_{\mathrm{Gauss}} = \frac{2\langle Q\rangle}{\mathrm{Var}(Q)}, \qquad G_{\mathrm{FT}} = \frac{\mathrm{Var}(Q)}{2\langle Q\rangle/\Delta\beta}.$$

They characterize the CLT crossover but are not substitutes for a directly sampled negative tail.

Provenance

Experiment repository commit: `e170205c5d9a3d9b2bfca3714a4447fa965a1e40`.
Production source commit: `1905cf4e606a4a7f4dd8930caa64bd4cc861e9d4`.
Frozen source SHA-256: `98e7f8f5f915c8ce02bd8aa10722025c09fd739184b981961692869c9356c0d3`.
Driven binary SHA-256: `4c4880d721733897d200f2601690da873f5b226aaa1013df23df2351c1dfe7d1`.
Equilibrium binary SHA-256: `5fdcd596f0b4da39357a7217fb8b475f14e7b381da08a1208351fccbc008080e`.

The four driven runs and the equilibrium amendment use $n = 10$, $\gamma = 0.1$, $dt = 5 \times 10^{-4}$, burn-in 500, base block duration 20, 8 SIMD batches, 16 lanes, 7,813 blocks per stream, and 1,000,064 blocks per case.

Exact production commands.

```
dbeta_0p027972  /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/bin/entropy_ft_affinity
↪  sample 6.5 5.5 10 8 500 20 7813 0.0005 2026090401 2
↪  /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/raw/dbeta_0p027972_n10 5
dbeta_0p057143  /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/bin/entropy_ft_affinity
↪  sample 7 5 10 8 500 20 7813 0.0005 2026090402 2
↪  /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/raw/dbeta_0p057143_n10 5
dbeta_0p125000  /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/bin/entropy_ft_affinity
↪  sample 8 4 10 8 500 20 7813 0.0005 2026090403 2
↪  /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/raw/dbeta_0p125000_n10 5
dbeta_0p222222  /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/bin/entropy_ft_affinity
↪  sample 9 3 10 8 500 20 7813 0.0005 2026090404 2
↪  /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/raw/dbeta_0p222222_n10 5
dbeta_0p000000
↪  /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/bin/entropy_ft_affinity_equilibrium
↪  sample 6 6 10 8 500 20 7813 0.0005 2026090405 2
↪  /Users/jayleenjiang/NLS_affinity_runtime/experiments/entropy_ft_affinity_sweep_2026-09-03/raw/dbeta_0p000000_n10 5
```

Input hashes.

Case	rows	SHA-256
`dbeta_0p027972`	1000064	`8b44be37ea8f566fb3862b6cd796dcfcf433be0888752babb876754c84d4c421`
`dbeta_0p057143`	1000064	`382496fcd194154b0f6d486f2d4da9de80d2d0791f94cb0f3e695d8e666a29cc`
`dbeta_0p125000`	1000064	`0404492717ce335c1e2eff7970d5d514435fabec9b91604fa92b7d4846cdf685`
`dbeta_0p222222`	1000064	`342f3783b40db98263cd685c8ab4e531c9d83842db6762b0bd0a06061642958d`
`dbeta_0p400000`	1000064	`a23806e82f5514a9c3375d10a6644946b6b57d0efe8452b3bbf397b6230f9929`
`dbeta_0p000000`	1000064	`7fd6827f12b42844de1142613af124702ca5535ba63f3c54ab71006b256aef4d`

Full absolute input paths are retained in `analysis/input_hashes.csv`.

Bootstrap-gate amendment and audit trail

The original all-resolved-time gate reported 0/1000 accepted intercepts for every case because sub-threshold raw acceptance was written as zero. The amended analysis leaves the production data, ΔQ, minimum count, minimum three pairs, and straddle-zero rule unchanged. Each t is bootstrapped independently on its frozen full-sample bin grid. Driven-case intercepts use only full-sample fits with $n_- \geq 500$ and $R^2 \geq 0.98$.

Case	old reported	old raw valid	old t set	amended t set	amended valid
$\Delta\beta = 0.027972$	0/1000	19/1000	20;40;80;160;320;640	20;40;80;160;320	1000/1000
$\Delta\beta = 0.057143$	0/1000	103/1000	20;40;80;160;320	20;40;80;160	1000/1000
$\Delta\beta = 0.125000$	0/1000	38/1000	20;40;80	20;40;80	1000/1000
$\Delta\beta = 0.222222$	0/1000	0/1000	20;40	20;40	no intercept
$\Delta\beta = 0.400000$	0/1000	0/1000	20	20	no intercept
$\Delta\beta = 0$	0/1000	29/1000	20;40;80;160;320;640	20;40;80;160;320;640	1000/1000

At equilibrium the literal $R^2 \geq 0.98$ set is empty because the exact reference is flat. The displayed equilibrium set therefore uses $n_- \geq 500$ without the inapplicable nonzero-signal R^2 gate.

Per-time stream-bootstrap intervals

The final column tests the finite-time interval against $\Delta\beta$ (zero at equilibrium), not the long-time verdict.

$\Delta\beta$	t	n_-	R^2	a_{fit}	CI low	CI high	accepted	in CI
0	20	499786	0.0214	1.6301242e-04	-9.3704041e-05	4.1582966e-04	1000/1000	yes
0	40	249282	1.0247e-04	-1.1067792e-05	-2.8364219e-04	2.4888232e-04	1000/1000	yes
0	80	124666	0.0010	3.8954372e-05	-2.1867434e-04	2.7890693e-04	1000/1000	yes
0	160	62425	0.0551	2.2666363e-04	-3.9092428e-05	4.9219777e-04	1000/1000	yes
0	320	31032	6.7076e-06	-2.8240726e-06	-2.6766063e-04	2.3607364e-04	1000/1000	yes
0	640	15544	0.0124	1.2068950e-04	-1.2930627e-04	3.9963393e-04	1000/1000	yes
0.027972	20	366513	0.9978	0.0272050	0.0268544	0.0275232	1000/1000	no
0.027972	40	155995	0.9979	0.0275799	0.0271995	0.0279071	1000/1000	no
0.027972	80	60756	0.9979	0.0278058	0.0273532	0.0281759	1000/1000	yes
0.027972	160	20136	0.9951	0.0270617	0.0264710	0.0275384	1000/1000	no
0.027972	320	5109	0.9944	0.0277764	0.0267290	0.0285637	1000/1000	yes
0.027972	640	764	0.9704	0.0303564	0.0267504	0.0341906	1000/1000	yes
0.057143	20	248420	0.9993	0.0548656	0.0544527	0.0552266	1000/1000	no
0.057143	40	82050	0.9991	0.0560181	0.0554428	0.0564510	1000/1000	no
0.057143	80	20447	0.9986	0.0565142	0.0555230	0.0573674	1000/1000	yes
0.057143	160	2997	0.9928	0.0561435	0.0538457	0.0578973	1000/1000	yes
0.057143	320	150	0.6020	0.0597810	-0.0326910	0.0923526	976/1000	yes
0.125000	20	88761	0.9995	0.1133690	0.1125397	0.1140623	1000/1000	no
0.125000	40	12744	0.9975	0.1196611	0.1175731	0.1213151	1000/1000	no
0.125000	80	661	0.9866	0.1246028	0.1116154	0.1342217	1000/1000	yes
0.222222	20	24798	0.9994	0.1761832	0.1740515	0.1778209	1000/1000	no
0.222222	40	985	0.9901	0.1914773	0.1772358	0.2077366	1000/1000	no
0.400000	20	7941	0.9985	0.2157315	0.2113351	0.2191951	1000/1000	no

Finite-time corrections remain visible. Rows failing the fixed negative-count or R^2 gate are reported here but excluded from the amended intercept.

Original per-window outputs retained for audit

The original bootstrap CI column below uses the fixed-full-window conditional bootstrap. All full-sample moments and slopes are unchanged.

dbeta_0p027972: $(T_L, T_R) = (6.5, 5.5)$, $\Delta\beta = 0.027972$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	8.9368	25.8599	0.1348	0.3798	366513	0.02720	[–,–]	0.9978	0.02673 / 1.0466
40	499968	17.8742	36.3277	0.1019	0.2084	155995	0.02758	[–,–]	0.9979	0.02709 / 1.0326
80	249984	35.7483	51.2908	0.0770	0.0995	60756	0.02781	[0.02736,0.02818]	0.9979	0.02718 / 1.0292
160	124928	71.4974	72.6266	0.0512	0.0388	20136	0.02706	[0.02645,0.02754]	0.9951	0.02711 / 1.0318

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
320	62464	142.9947	102.6171	0.0433	0.0109	5109	0.02778	[–,–]	0.9944	0.02716 / 1.0299
640	31232	285.9894	145.9539	0.0505	0.0103	764	0.03036	[–,–]	0.9704	0.02685 / 1.0418

dbeta_0p057143: $(T_L, T_R) = (7.0, 5.0)$, $\Delta\beta = 0.057143$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	17.7373	26.2279	0.2518	0.4064	248420	0.05487	[0.05446,0.05523]	0.9993	0.05157 / 1.1081
40	499968	35.4752	36.8119	0.1869	0.2167	82050	0.05602	[–,–]	0.9991	0.05236 / 1.0914
80	249984	70.9503	51.9375	0.1401	0.0936	20447	0.05651	[–,–]	0.9986	0.05260 / 1.0863
160	124928	141.8966	73.3519	0.0951	0.0409	2997	0.05614	[–,–]	0.9928	0.05274 / 1.0834
320	62464	283.7931	103.2913	0.0737	0.0348	150	0.05978	[–,–]	0.6020	0.05320 / 1.0741
640	31232	567.5862	145.5179	0.0377	0.0130	1	UNRES.	–	–	0.05361 / 1.0659

dbeta_0p125000: $(T_L, T_R) = (8.0, 4.0)$, $\Delta\beta = 0.125000$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	34.8992	27.7171	0.4529	0.5303	88761	0.11337	[–,–]	0.9995	0.09086 / 1.3758
40	499968	69.7973	38.8933	0.3447	0.2789	12744	0.11966	[–,–]	0.9975	0.09228 / 1.3545
80	249984	139.5947	54.7524	0.2520	0.1553	661	0.12460	[–,–]	0.9866	0.09313 / 1.3422
160	124928	279.1903	77.2501	0.1806	0.0808	2	UNRES.	–	–	0.09357 / 1.3359
320	62464	558.3805	109.0295	0.1323	0.0777	0	UNRES.	–	–	0.09394 / 1.3306
640	31232	1116.7611	153.4323	0.1020	0.0404	0	UNRES.	–	–	0.09488 / 1.3175

dbeta_0p222222: $(T_L, T_R) = (9.0, 3.0)$, $\Delta\beta = 0.222222$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	50.8282	29.9660	0.5614	0.5940	24798	0.17618	[0.17407,0.17782]	0.9994	0.11321 / 1.9630
40	499968	101.6576	42.1593	0.4311	0.3267	985	0.19148	[–,–]	0.9901	0.11439 / 1.9427
80	249984	203.3151	59.3516	0.3233	0.1607	3	UNRES.	–	–	0.11543 / 1.9251
160	124928	406.6254	83.9248	0.2323	0.0889	0	UNRES.	–	–	0.11546 / 1.9246
320	62464	813.2508	119.0787	0.1714	0.0688	0	UNRES.	–	–	0.11471 / 1.9373
640	31232	1626.5015	168.5450	0.0984	0.0109	0	UNRES.	–	–	0.11451 / 1.9406

dbeta_0p400000: $(T_L, T_R) = (10.0, 2.0)$, $\Delta\beta = 0.400000$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	64.2424	32.6857	0.6006	0.6213	7941	0.21573	[–,–]	0.9985	0.12026 / 3.3260
40	499968	128.4849	45.9342	0.4603	0.3523	80	UNRES.	–	–	0.12179 / 3.2844
80	249984	256.9699	64.7162	0.3377	0.1853	0	UNRES.	–	–	0.12271 / 3.2597
160	124928	513.9476	91.2279	0.2361	0.0836	0	UNRES.	–	–	0.12351 / 3.2387
320	62464	1027.8951	128.6532	0.1634	0.0240	0	UNRES.	–	–	0.12420 / 3.2205
640	31232	2055.7903	182.2058	0.1061	0.0165	0	UNRES.	–	–	0.12385 / 3.2298

dbeta_0p000000: $(T_L, T_R) = (6.0, 6.0)$, $\Delta\beta = 0$

t	N	mean	std	skew	ex.kurt	n_-	a_{fit}	boot.95% CI	R^2	a_G / G_{FT}
20	1000064	0.0306	25.7283	-0.0027	0.3783	499786	1.63012e-04	[-8.97958e-05,4.18259e-04]	0.0214	9.24910e-05 / –
40	499968	0.0610	36.1107	-0.0068	0.2064	249282	-1.10678e-05	[–,–]	1.0247e-04	9.34879e-05 / –
80	249984	0.1219	50.8838	-0.0029	0.1190	124666	3.89544e-05	[–,–]	0.0010	9.41668e-05 / –
160	124928	0.2379	71.8208	-3.2658e-04	0.0612	62425	2.26664e-04	[–,–]	0.0551	9.22287e-05 / –
320	62464	0.4757	101.3193	-0.0103	0.0283	31032	-2.82407e-06	[–,–]	6.7076e-06	9.26856e-05 / –
640	31232	0.9515	143.2340	0.0089	-0.0130	15544	1.20689e-04	[–,–]	0.0124	9.27544e-05 / –

Amended reliable-time extrapolation

Case	$\Delta\beta$	reliable t	a_∞	95% CI	$a_\infty/\Delta\beta$	ratio CI	status
dbeta_0p027972	0.027972	20;40;80;160;320	0.02761	[0.02700,0.02805]	0.9871	[0.9652,1.0027]	CONSISTENT_WITH_FT
dbeta_0p057143	0.057143	20;40;80;160	0.05668	[0.05503,0.05787]	0.9919	[0.9631,1.0127]	CONSISTENT_WITH_FT
dbeta_0p125000	0.125000	20;40;80	0.12775	[0.11451,0.13764]	1.0220	[0.9161,1.1011]	CONSISTENT_WITH_FT
dbeta_0p222222	0.222222	20;40	–	[–,–]	–	[–,–]	UNRESOLVED
dbeta_0p400000	0.400000	20	–	[–,–]	–	[–,–]	UNRESOLVED
dbeta_0p000000	0	20;40;80;160;320;640	7.80343e-05	[-1.17836e-04,2.58983e-04]	–	[–,–]	CONSISTENT_WITH_EQUILIBRIUM

The amended intercept accepts 1000/1000 replicates for the first three driven rows and equilibrium. Regression R^2 is retained as a diagnostic; the weakest-drive linear $1/t$ extrapolation is noisy.

Crossover summary

$\Delta\beta$	$a_\infty/\Delta\beta$	ratio CI	$a_G/\Delta\beta$	$G_{FT}(640)$	skew(160)	ex.kurt(160)	$n_-(160)$	status
0	–	[–,–]	–	–	-3.2658e-04	0.0612	62425	CONSISTENT_WITH_EQUILIBRIUM
0.027972	0.9871	[0.9652,1.0027]	0.9599	1.0418	0.0512	0.0388	20136	CONSISTENT_WITH_FT
0.057143	0.9919	[0.9631,1.0127]	0.9310	1.0659	0.0951	0.0409	2997	CONSISTENT_WITH_FT
0.125000	1.0220	[0.9161,1.1011]	0.7450	1.3175	0.1806	0.0808	2	CONSISTENT_WITH_FT
0.222222	–	[–,–]	0.5147	1.9406	0.2323	0.0889	0	UNRESOLVED
0.400000	–	[–,–]	0.3007	3.2298	0.2361	0.0836	0	UNRESOLVED

Figures

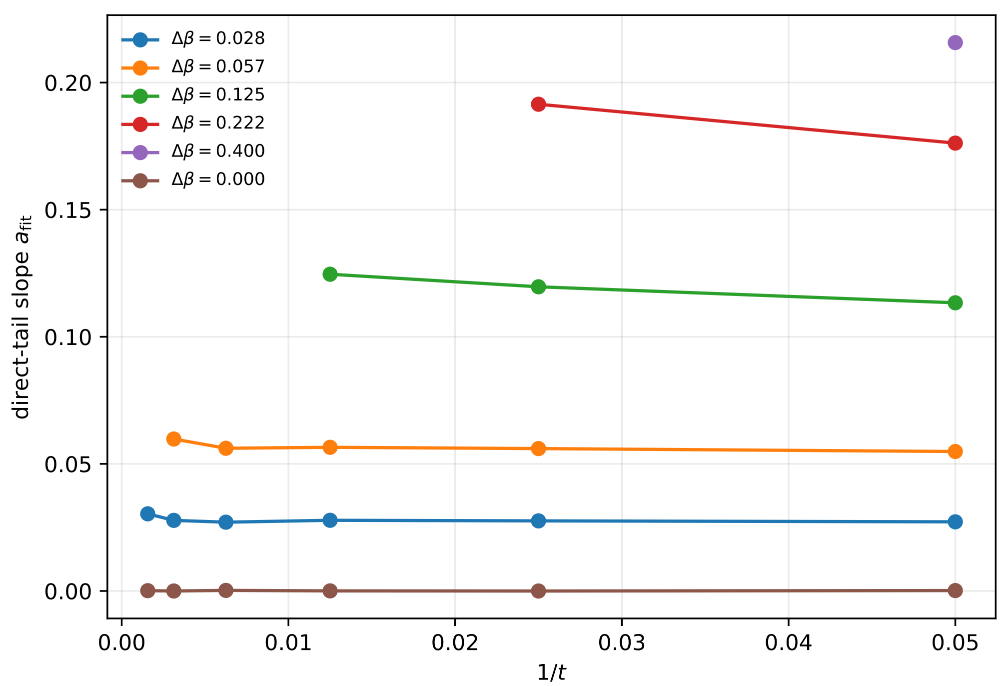

Figure 1: Direct matched-bin slopes against inverse window duration. The amended intercept uses only the reliable-time subsets stated above.

Equilibrium row and claim boundary

The added $(6, 6)$ production uses the same $n = 10$ dynamics, integration step, burn-in, stream count, and block count as the driven cases. At $\Delta\beta = 0$, ratios that divide by $\Delta\beta$ and $G_{\rm FT}$ are undefined and are reported as dashes. The

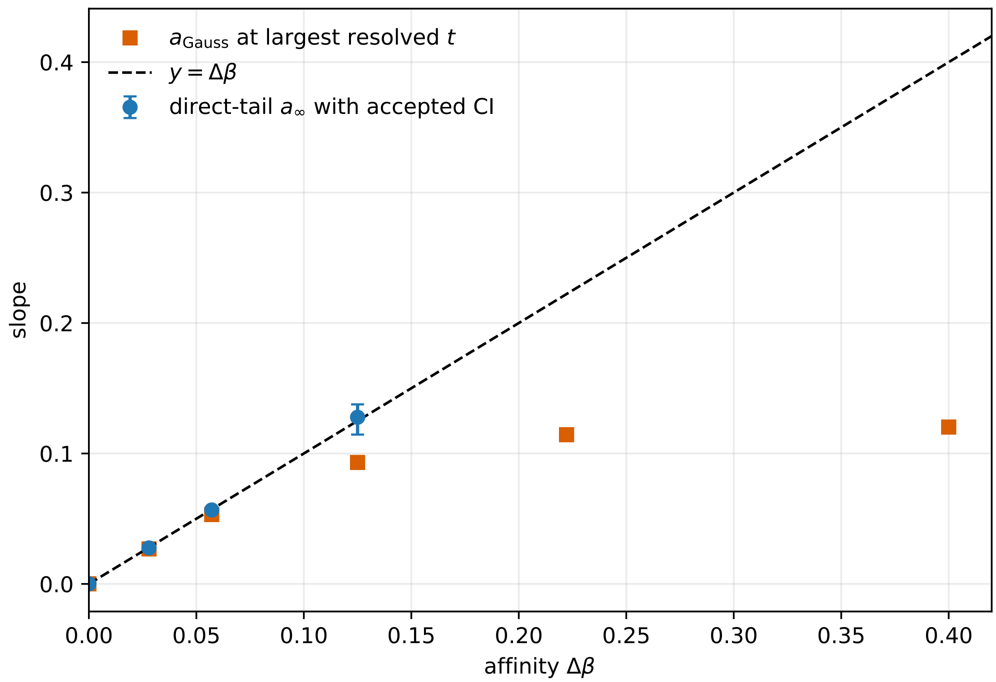

Figure 2: Amended reliable-time direct-tail intercepts and Gaussian bulk slopes against affinity.

direct asymmetry-slope reference is zero; its extrapolated confidence interval is tested against zero rather than against a ratio of one.

The complete raw positive/negative counts used by every fit are in `analysis/symmetric_bin_raw_counts.csv`; all per-window statistics are in `analysis/window_summary.csv`. Old and amended gates are side by side in `analysis/bootstrap_gate_comparison.csv`; pre-amendment artifacts are under `pre_gate_respec_2026-09-04/`. A case labelled `CONSISTENT_WITH_FT` means only that the amended reliable-time extrapolated interval includes the reference. It is not a proof. A case whose interval excludes the reference is labelled `FAIL`; insufficient raw two-tail support remains `UNRESOLVED`. The label `CONSISTENT_WITH_EQUILIBRIUM` has the corresponding limited meaning for the zero-affinity control.

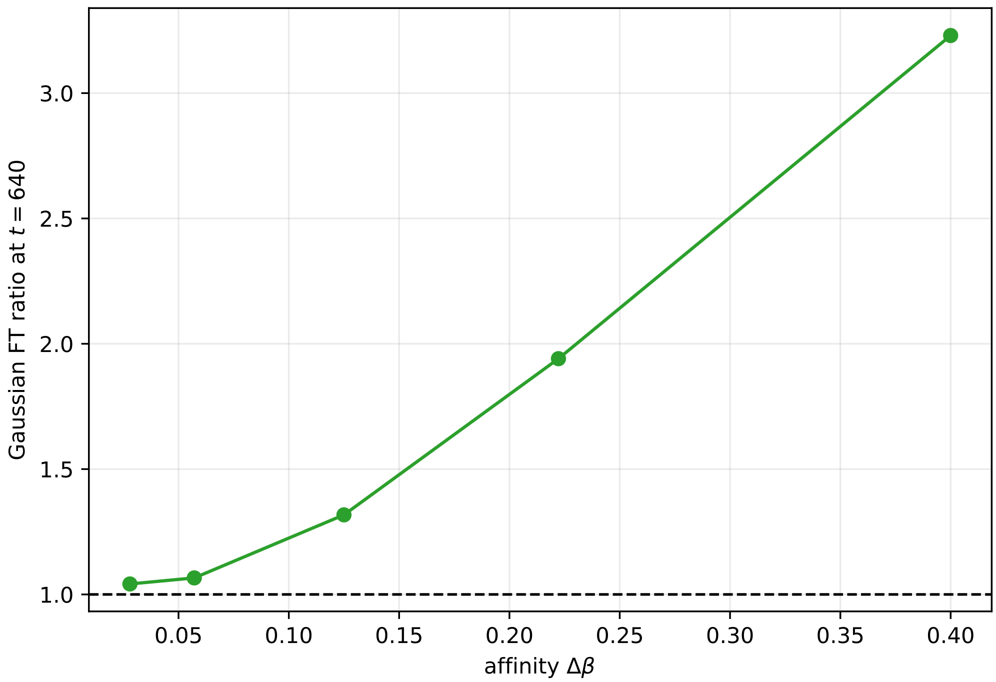

Figure 3: Gaussian FT ratio at $t = 640$ across affinity.